

Channel Performance and AB Test Analysis for Flowerpot and Co.

Analysis by Nitesh Talukdar

Problem statement

Flowerpot and Co. is a fictional online store that sells homeware, clothing, beauty products, electronics and outdoor goods in five countries. Between January 2021 and December 2023 it brought visitors to its website through nine marketing channels, including search, social media, display adverts, affiliates and email. It also ran three tests on its email campaigns. Every channel costs money or effort, and a marketing manager needs to know which ones turn visitors into customers and which ones do not. A manager running this business would need clear answers to four questions.

Channel performance. Which channels bring visitors who buy, and how much revenue does each visit earn.

Trends. Whether sessions, conversion and revenue are moving up or down from one month to the next, and how much of that is the usual holiday season.

Email. How well email campaigns do, measured by how many people open, click, buy or unsubscribe, and whether sending more email helps.

Tests. Whether the changes tried in three email tests really worked, or whether the gap could be luck.

The goal of this project is to answer these questions with the data, show the limits of each answer, and turn the results into actions a marketing manager could take. The answers sit in a two page Power BI dashboard that a manager can read without opening any data.

The dataset

The data is a synthetic online store dataset, free to use and published on Kaggle. A Python script with a fixed seed made it to look like the customer, website and order data of a real store, so it holds no real customers or companies. It has 11 tables and about 3.5 million rows, from January 2021 to December 2023. Orders are written in five currencies, which are US dollars, British pounds, euros, Canadian dollars and Australian dollars. The dashboard uses eight of the tables.

What it holds	Rows in the raw file
One row per customer account, used here to find internal test accounts	45,560
One row per marketing campaign, with channel, dates and spend	107 lines, 102 campaigns
The three email tests, with the hypothesis and the two versions	3
Monthly rates for four currencies against the US dollar	36 months
One row per email sent, with the test version and delivery status	965,024
Opens, clicks, unsubscribes and spam complaints	431,858
One row per website visit, with the page the visitor landed on and where they came from	606,384
One row per order, with status, currency, total and refund	18,571

Three tables are not used. Products has 600 rows, order items has 28,633 and website events has 1,409,685. They could feed a product page and a funnel page later.

Tools and skills

I used three tools, one after the other. SQL checked the raw data and removed what should not be counted. Power Query fixed the spellings, dates and money. Power BI and DAX turned the clean tables into measures and a report, and each step left a record that someone else could check.

SQL. I loaded the 11 files into a SQLite database and used SQL to check the data before using it. A profiling file asks one question at a time, such as how many ways an event is written, how many rows are exact copies and which orders have no matching visit. A second file creates six views that fix the problems SQL handles well, and the raw tables stay untouched, so every fix can be checked and changed. A third file counts the rows after every filter and compares them with the expected results.

Skills shown. Grouping and counting, joins that find missing links, subqueries, common table expressions, case statements, careful handling of blank values, turning many event rows into one row per email, reshaping exchange rates by stacking queries, and validation queries.

Value to the business. Numbers that can be trusted, because every removal is counted and explained.

Power Query. I used it for the cleaning that is easier to do step by step, before the data reached the model.

Skills shown. Mapping many spellings to one value, writing a small function that reads money written in five formats, reading tracking tags out of web addresses, parsing dates with a fallback format, joining orders to exchange rates and to website visits, and grouping.

Value to the business. Every order ends up in one currency and every visit has a channel, which is what makes channel comparisons fair.

Power BI and DAX. I built a data model with two fact tables, visits and emails, that share a date table and a campaign table, and then built a two page report on top of it, with the same layout and colours on both pages.

Skills shown. Relationships and a marked date table; DAX for ratios, month over month change with time functions, a three month average, ranking, a two group z test with a confidence range, and text measures that write a headline for the selected month or test. On the report side, cards with change labels and conditional colours, a table with data bars, line charts, an error bar chart, a month picker and a test picker.

Value to the business. A manager gets the answer in one sentence and can then check the detail underneath.

Challenges and assumptions

The raw data could not be used as it was. Each problem below needed a decision, and the SQL files record the count for each one.

Test accounts and test traffic. 20 internal test accounts owned 656 emails, 365 visits and 29 orders, and 4,215 visits came from a staging copy of the website. These are not customers, so I removed them. One view holds the list of test accounts, and every other view leaves those accounts out.

Exact duplicates. 5,756 email rows and 56 orders were exact copies, so I kept one of each. After all removals, email sends went from 965,024 rows to 958,618, sessions from 606,384 to 601,804, and orders from 18,571 to 18,486.

Visitors who are not logged in. 235,826 visits have no customer number. A simple filter that leaves out test accounts would have dropped every one of them, because a blank value never passes a not in test. I wrote the filter to keep them, so visit counts stay correct.

Event types written 13 ways. An open was written as open, Open, OPEN or opened, and clicks and unsubscribes had the same problem. Counting only the plain spelling would have put the open rate near 19 percent, when the true figure is about 27 percent. I grouped every spelling, ignoring capitals and spaces.

Clicks with no open. 2,076 of 25,995 clicked emails had no open recorded, because some email apps block the tracking image. A click proves the email was opened, so I counted these as opened.

Emails with no events. 701,099 emails had no open, click or other event. I kept them as not opened. Dropping them would have made the open rate far too high.

A campaign file with title lines. The export had three title lines, a header and a total line, so it loaded as 107 rows. A view gives it proper column names and keeps only the 102 real campaigns.

Many spellings of the same value. The test version was written 10 ways, delivery status 9 ways, order status 11 ways, currency 11 ways, campaign channel 12 ways, the tracking medium 12 ways and device 11 ways. I mapped each to a short list in Power Query.

Money in five formats. Order totals and refunds were text. Dollar, pound and similar amounts used a comma for thousands. Euro amounts used a dot for thousands and a comma for decimals, so 1.204,10 means one thousand two hundred and four euros ten. 2,110 orders were in euros. I chose how to read each amount from the currency code, not from the symbol, and converted everything to US dollars with the exchange rate of the order month.

Rules for order status. Net revenue is the order total minus refunds. The 281 cancelled orders count as no order and no revenue. The 615 refunded and 380 partly refunded orders still count as orders, because the order happened, but their revenue is reduced.

The channel is hidden in the web address. The visits table has no channel column. The tracking tags in the landing page address carry it, written 12 ways. For the 350,821 visits with no tags, I used the site the visitor came from. The order of the rules matters. 5,186 email visits arrive with a Gmail referrer, so tags are checked first. Otherwise they would be counted as organic search.

Email orders. An order counts as an email order only when it was placed in a visit that started from a click on that email. This follows the rule in the data dictionary. It gives 25,950 email visits and 1,608 email orders.

Month of an order. Revenue is dated by the visit. Four orders were placed in a different month from their visit, so a month can differ by a few orders from a view dated by order. Yearly totals match.

Noise in small groups. Monthly figures for small channels swing a lot. Email had as few as 154 visits in one month, and its revenue per visit moved between 1.77 and 7.07 US dollars. For the trend chart I used a three month average and showed only the four largest channels.

Campaign spend. Spend was written in three formats, some values were rounded, such as 1.7k, and email spend is blank because it is not tracked. Total spend comes out about 38 US dollars above the documented figure. Spend is in the model but is not used in the dashboard yet.

The same customers in many emails. In each test the same customers received several emails. The statistics treat each email as separate, so the confidence ranges are a little narrower than they should be.

After cleaning, I checked the numbers against the expected results in the dataset documents. Total visits, the conversion rate, net revenue, the email open rate and the group sizes of all three tests matched.

Calculations and why they matter

Conversion rate

Orders divided by sessions, shown as a percent. It is the plainest test of whether a channel brings people who buy. A channel with many visits and a low rate takes effort and gives few orders. The overall rate is 3.03 percent.

Revenue per session

Net revenue divided by sessions. It joins the conversion rate and the size of the order in one number, so channels of different sizes can be compared fairly. The overall figure is 2.39 US dollars.

Net revenue

The order total minus refunds, in US dollars, with cancelled orders at zero. A manager needs this figure, not the sum of invoices, because refunds and cancellations are not real income.

Revenue share

Each channel revenue divided by total revenue. It shows which channels matter most, so that a high rate on a very small channel does not pull attention away from the large ones.

Email rates

The open rate, click rate and unsubscribe rate use delivered emails as the base, not emails sent, because a bounced email cannot be opened. The email conversion rate is orders credited to email divided by delivered emails. Revenue per send is email revenue divided by emails sent.

Month over month change

The chosen month is compared with the month before. For counts and money I show the percent change. For rates I show the change in percentage points, because a percent change of a percent confuses readers. For example, the open rate fell 2.51 points, not 9 percent.

Three month average

Revenue per session over a month and the two before it, found by adding revenue and sessions first and then dividing. It removes the month to month noise that makes small channels hard to read.

Lift, confidence range and verdict

Lift is how much better version B did than version A, the control, as a share of A. The 95 percent range is where the true difference probably lies. If the whole range is above zero, B is better. If it crosses zero, I cannot tell yet. The p value comes from a two group z test and is the chance of seeing a gap this big if the two versions were really equal. A result is Significant when the p value is below 0.05. A test that is still running and has not reached that mark is shown as Not yet significant, since more data may change the answer.

Findings

1. Email is the best channel for each visit, but it is a small part of the business. Email visits converted at 6.20 percent against 3.03 percent for the whole site, and each visit earned 4.70 US dollars against 2.39. Email brought only 25,950 of 601,804 visits, about 4 percent, and 122,074 of 1.44 million US dollars of revenue, about 8.5 percent.

2. Four channels bring in most of the revenue, and three convert poorly. Organic search, direct, paid search and email together bring in 85.5 percent of net revenue. Display converts at 0.59 percent and earns 0.47 US dollars a visit. Organic social converts at 0.85 percent and earns 0.70, and paid social at 1.02 percent and 0.80. Paid social and display are paid channels, so their visits have a cost, and they are the first place to review budgets.

Channel	Visits	Orders	Conversion in percent	Net revenue in US dollars	Revenue per visit in US dollars	Share of revenue in percent
Organic search	190,026	5,700	3.00	449,275	2.36	31.2
Direct	108,820	4,508	4.14	355,083	3.26	24.7
Paid search	115,149	3,783	3.29	304,352	2.64	21.1
Email	25,950	1,608	6.20	122,074	4.70	8.5
Affiliate	23,130	948	4.10	77,724	3.36	5.4
Paid social	63,643	651	1.02	50,905	0.80	3.5
Referral	23,314	626	2.69	49,752	2.13	3.5
Organic social	28,661	245	0.85	20,084	0.70	1.4
Display	23,111	136	0.59	10,927	0.47	0.8
All channels	601,804	18,205	3.03	1,440,175	2.39	100.0

3. Growth came from more visits, not from better conversion. Visits rose 73 percent from 2021 to 2023 and net revenue rose 78 percent, but the conversion rate only moved from 2.98 to 3.08 percent, and revenue per visit stayed near 2.40 US dollars. More traffic is paying off, but the site is not yet turning visitors into buyers any better.

Year	Visits	Orders	Conversion in percent	Net revenue in US dollars	Revenue per visit in US dollars
2021	149,128	4,441	2.98	353,838	2.37
2022	194,269	5,817	2.99	454,906	2.34
2023	258,407	7,947	3.08	631,432	2.44

4. November and December lift every number, and December 2023 shows the pattern. In 2023, visits rose from 22,340 in October to 33,557 in November, and conversion rose from 2.68 to 3.72 percent. In December 2023 visits fell 3.6 percent, conversion fell 0.24 points to 3.47 percent, and net revenue fell 2.1 percent to 89,529 US dollars. Email sends rose 157 percent to 131,599 for the holiday campaigns, while the open rate fell 2.51 points to 26.65 percent. Changes in these two months should be judged against the same months of earlier years, not against October.

5. Sending much more email did not raise what each email earns. Emails sent rose almost four times from 2021 to 2023, but email revenue rose 3.7 times, so revenue per send slipped from 0.136 to 0.128 US dollars, and the open rate drifted down from 27.53 to 26.93 percent. Sending more is not a plan on its own. Who gets each email and what it says still need testing.

Year	Emails sent	Open rate in percent	Click rate in percent	Orders from email	Email revenue in US dollars	Revenue per send in US dollars
2021	143,413	27.53	2.76	246	19,562	0.136
2022	244,737	27.29	2.67	417	29,213	0.119
2023	570,468	26.93	2.76	945	73,298	0.128

6. Two of the three email tests produced a real win. In every test, version B is the new idea and version A is the control. Test T01 put the first name in the subject line and raised the open rate by 11.6 percent. Test T02 used lifestyle photos in place of a product grid and raised the click rate by 25.0 percent. Both ranges sit fully above zero. Test T03 compared 15 percent off with free shipping. Its conversion lift of 28.1 percent looks large, but it rests on only 74 and 96 orders and its range crosses zero, so it is Not yet significant. It was still running when the data ends, with a planned end on 16 January 2024.

Test	What changed in version B	Main measure	A rate in percent	B rate in percent	Lift in percent	95 percent range in points	p value	Verdict
T01	First name in the subject line	Open rate	26.14	29.17	11.6	2.41 to 3.65	0.0000	Significant
T02	Lifestyle photos in place of a product grid	Click rate	2.60	3.25	25.0	0.43 to 0.87	0.0000	Significant
T03	15 percent off in place of free shipping	Conversion rate	0.14	0.18	28.1	minus 0.01 to plus 0.09	0.1081	Not yet significant

Two details are worth watching. In T01 the click rate also rose by 10.5 percent, with a p value of 0.0198. The unsubscribe rate in T01 rose from 0.17 to 0.24 percent, with a p value of 0.0547, which is close to the line, so personalised subject lines should be watched for unsubscribes. In T02 there was no clear effect on opens, orders or unsubscribes.

What I would do

1. Use the first name in subject lines and the lifestyle layout in regular campaigns, and keep watching the unsubscribe rate.
2. Wait for the end of test T03 before choosing between a discount and free shipping.
3. Review the budgets of paid social and display against their low revenue per visit.
4. Judge November and December against the same months of earlier years.
5. Test who gets each email and what it says before sending more email.

The dashboard

The dashboard puts the main numbers on two pages for a manager who does not want to read the detail. On page one, a month dropdown at the top right sets the month, and a headline sentence under the title says what changed. The left half shows the website, and the right half shows email. Each card shows the change from the month before, in green when it is better and red when it is worse. For unsubscribes a fall is green. A table ranks the nine channels, with a column for revenue share and a column for the change in conversion. The trend charts show every month and do not follow the dropdown, so they always show the full history.

Number shown	What it tells you	Value for December 2023
Sessions	Visits to the website in the month	32,349
Conversion rate	Share of visits that placed an order	3.47 percent
Revenue per session	Net revenue earned by each visit	2.77 US dollars
Net revenue	Order totals after refunds and cancellations	89,529 US dollars
Sends	Emails sent in the month	131,599
Open rate	Share of delivered emails that were opened	26.65 percent
Click rate	Share of delivered emails that were clicked	2.65 percent
Email conversion rate	Share of delivered emails that led to an order	0.186 percent
Revenue per send	Email revenue for each email sent	0.138 US dollars
Unsubscribe rate	Share of delivered emails that led to an unsubscribe	0.187 percent

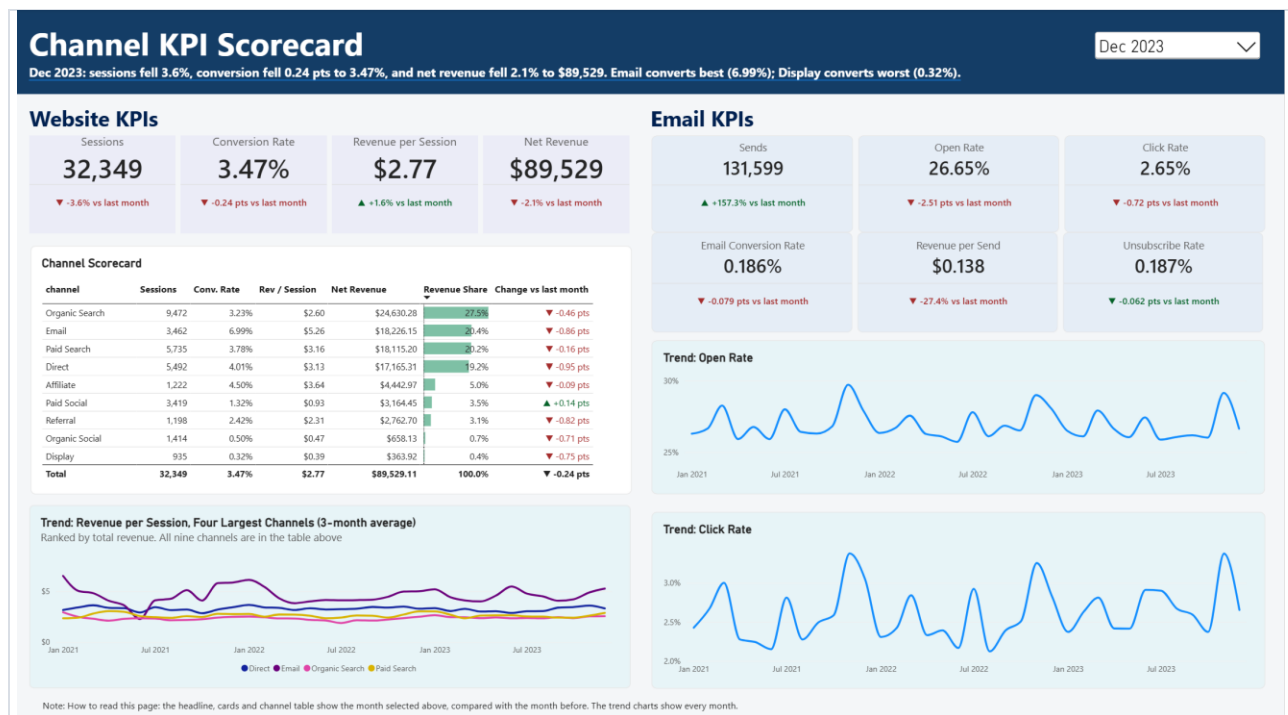


Figure 1 Page one of the dashboard, set to December 2023

Page two reads out one test at a time. Three tiles at the top choose the test. A headline sentence states the main result in plain words. A details strip shows the hypothesis, the two versions, the dates and the status. The results table shows, for each of four measures, the size of each group, the rates, the lift, the 95 percent range, the p value and the verdict. Under it a chart shows the difference with its range. A dashed line marks no difference, and a range that crosses it means the result is not clear yet. Green bars are Significant, grey bars are Not significant and amber bars are Not yet significant.

A/B Test Readout

Did the variant beat the control? Pick a test below.

T01 - Personalised subject line

T02 - Lifestyle vs product-grid email layout

T03 - Free shipping vs 15% off offer

Variant B raised open rate by 11.6% compared with A. 95% range of the difference: +2.41 to +3.65 points. Significant.

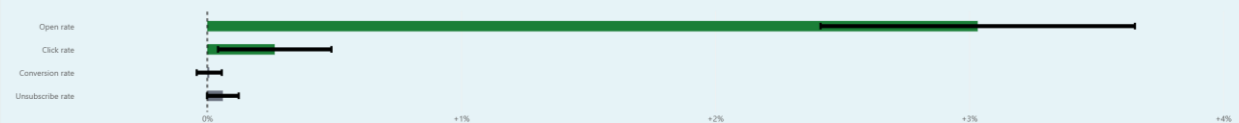
Test Details

Hypothesis	Control (A)	Variant (B)	Primary metric	Start date	Planned end date	Status
Adding the first name to the subject line will raise the open rate.	Subject without name	Subject with first name	Open rate	7 March 2023	4 April 2023	Completed

Test Results

Metric	Control size	Variant size	Control rate	Variant rate	Lift	CI low	CI high	P-value	Result
Open rate	39,722	40,418	26.14%	29.17%	+11.6%	+2.41%	+3.65%	0.0000	Significant
Click rate	39,722	40,418	2.54%	2.81%	+10.5%	+0.04%	+0.49%	0.0198	Significant
Conversion rate	39,722	40,418	0.12%	0.13%	+6.3%	-0.04%	+0.06%	0.7577	Not significant
Unsubscribe rate	39,722	40,418	0.17%	0.24%	+35.3%	-0.00%	+0.12%	0.0547	Not significant

Difference between Variant B and Control, with 95% confidence interval (percentage points)
Dashed line = no difference. A whisker that crosses it means the result is not clear yet.



Note: Unit = one delivered email. The same customers received several emails in each test, so results look slightly more certain than they should. T03 was still running when the data ends, so its result is interim.

Figure 2 Page two of the dashboard, showing test T01

The model behind the dashboard has two fact tables, website visits and email sends, that share a date table and a campaign table. The test details link through the campaign table.

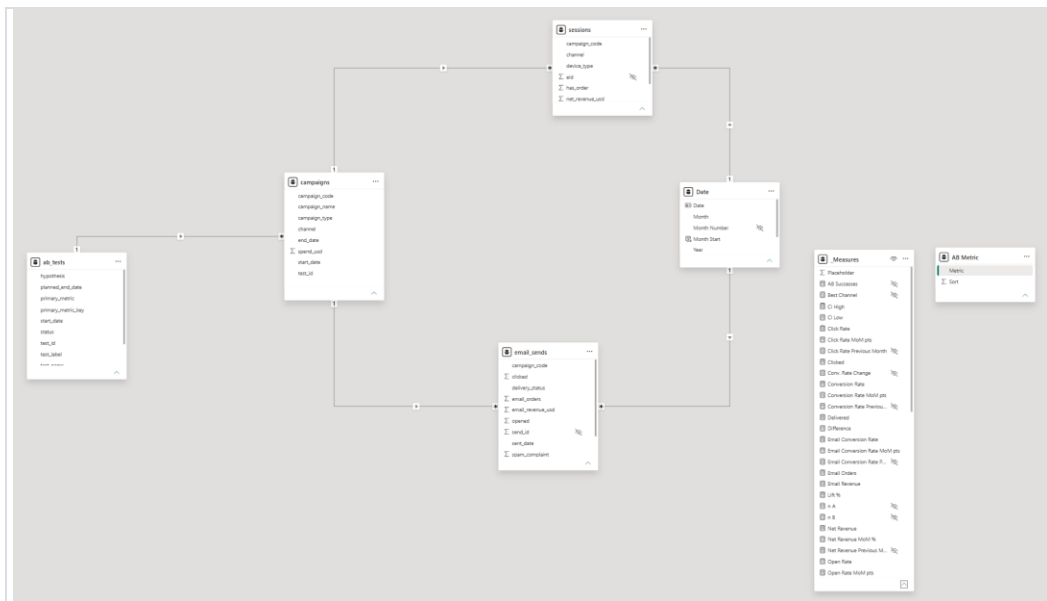


Figure 3 The data model

Limits

The data is synthetic, so the findings show how the method works and say little about real customers. Channel comparisons are not proof that moving budget between channels would give the same results, because visitors who arrive by different channels are different people. Each order is credited to the visit it was placed in, so a shopper who clicked an email and bought later through a direct visit counts as direct. There is no email cost in the data, and the spend for paid channels is not yet in the dashboard, so the project does not show return on ad spend.

In the tests, the same customers received several emails, so the confidence ranges are a little too narrow. Each test checks four measures, so across all three tests one result could look significant by chance. The main measure of each test gives the verdict. Test T03 was not finished. Some spend values in the source are rounded. Revenue is dated by the visit and not by the order. Repeat purchases, customer value, products and the path through the site are not covered.